



REVIEW ARTICLE

Graphene and functionalised nano-derivatives in water purification: A review

Prabu Padanillay Chidambaram¹, Muppidi Teja Gangadhar Reddy², Sengottiyar Priyatharshini^{2*}, Raja Kalimuthu³, Kumaraperumal Ramalingam⁴, Ragunath Kaliaperumal¹, Nivas Raj Moorthi⁴, Karthikeyan Ganesan², Rakesh S S² & Annie Josephine J²

¹Centre for Water and Geospatial Studies, Tamil Nadu Agricultural University, Coimbatore 641 003, Tamil Nadu, India

²Department of Environmental Sciences, Tamil Nadu Agricultural University, Coimbatore 641 003, Tamil Nadu, India

³Centre for Agricultural Nanotechnology, Tamil Nadu Agricultural University, Coimbatore 641 003, Tamil Nadu, India

⁴Department of Remote Sensing and GIS, Tamil Nadu Agricultural University, Coimbatore 641 003, Tamil Nadu, India

*Correspondence email - priya18ens@gmail.com

Received: 20 December 2024; Accepted: 29 November 2025; Available online: Version 1.0: 15 September 2026

Cite this article: Prabu PC, Muppidi TGR, Sengottiyar P, Raja K, Kumaraperumal R, Ragunath K, Nivas RM, Karthikeyan G, Rakesh SS, Annie JJ. Graphene and functionalised nano-derivatives in water purification: A review. Plant Science Today. 2026;13(sp5):01–10. <https://doi.org/10.14719/pst.6819>

Abstract

The growing demand for clean and safe water has intensified research on advanced materials capable of addressing complex wastewater contaminants. Graphene and its functionalised nano-derivatives have emerged as versatile materials in water purification due to their exceptional physicochemical properties, including high surface area, tunable surface chemistry, superior mechanical strength and excellent electronic conductivity. The high surface area, mechanical strength and chemical stability of graphene-based materials enhance their adsorption capacity and effectiveness in removing heavy metals and trapping organic contaminants, such as pharmaceuticals and benzene-related chemicals, from wastewater. The π - π stacking, electrostatic interactions and surface functional groups play critical roles in promoting adsorption processes and enhancing the efficacy of graphene-based materials for contaminant removal from wastewater streams. This review summarises recent advancements in the use of graphene and its derivatives for wastewater treatment, providing insights into the underlying mechanisms governing contaminant removal and the potential of these nanocarbon materials for global water pollution mitigation.

Keywords: adsorption; functional groups; graphene; heavy metals; organic pollutants

Introduction

Traditional wastewater treatment methods often face challenges regarding effectiveness, cost, and environmental impact (1). Graphene and its nano-derivatives have emerged as highly promising materials for advanced wastewater treatment due to their excellent physicochemical properties, including large surface area, tunable surface chemistry and notable electrical and thermal stability (2). Among the diverse nanomaterials explored, nanocarbons derived from naturally available materials such as graphite, pre-carbonised materials and agro-waste have garnered considerable attention. These nanocarbons, including graphene, graphene oxide (GO), reduced graphene oxide (rGO), graphene nanoplatelets (GNPs), graphene nanoribbons (GNRs), graphene quantum dots (GQDs), graphene aerogel (GA) or graphene foam, carbon nanotubes (CNTs), fullerenes and carbon nanofibers (CNFs), possess unique properties such as high surface area, tunable pore structure, and excellent adsorption capacity, making them ideal candidates for wastewater remediation (Fig. 1) (3). Graphene is considered one of the most ground-breaking materials of the twenty-first century. The single two-dimensional sheet of graphite, known as graphene, is the precursor of all nano-allotropes (4). This monolayer of graphite is only one atom thick (~0.34 nm), yet its

lateral dimension may be several orders of magnitude larger (5, 6). Graphene is currently used in a wide range of scientific and engineering domains, as illustrated in Fig. 2.

Graphene and its properties

Graphene, in which each carbon atom is sp^2 -hybridised, is formed by the hybridisation of ones, p_x and one p_y orbitals and each atom forms three σ -bonds with neighbouring carbon atoms. These carbon atoms are covalently bonded in a hexagonal lattice and they are all sp^2 -hybridised, resulting in one delocalised electron per carbon atom. This free electron is associated with the p_z orbital, which forms the π -bonding system located above and below the basal plane. Despite being more than 200 times stronger than steel, graphene is remarkably flexible (7). It is also highly durable while remaining extremely lightweight. It is highly transparent, making it one of the thinnest materials known. Graphene was first observed under an electron microscope was in 1962. With a theoretical surface area of around $2630 \text{ m}^2 \text{ g}^{-1}$, graphene is regarded as a material with an exceptionally large surface area of approximately $1000 \text{ m}^2 \text{ g}^{-1}$ (8). It is characterised by high adsorption efficiency, nanoscale dimensions and large specific surface area. For instance, π - π stacking, electrostatic interactions and van der Waals interactions forces

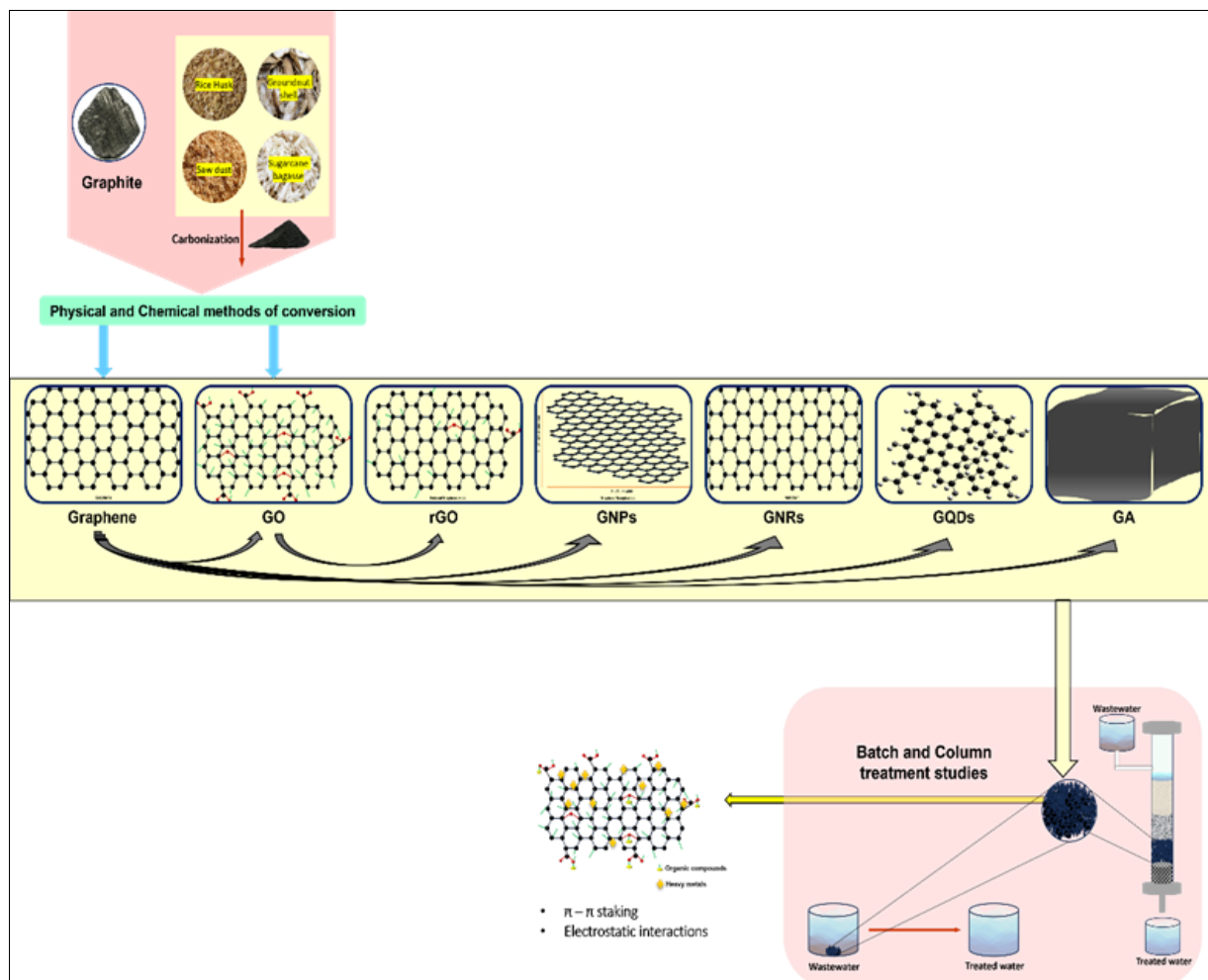


Fig. 1. Graphene and its nano derivatives used in advanced wastewater treatment.

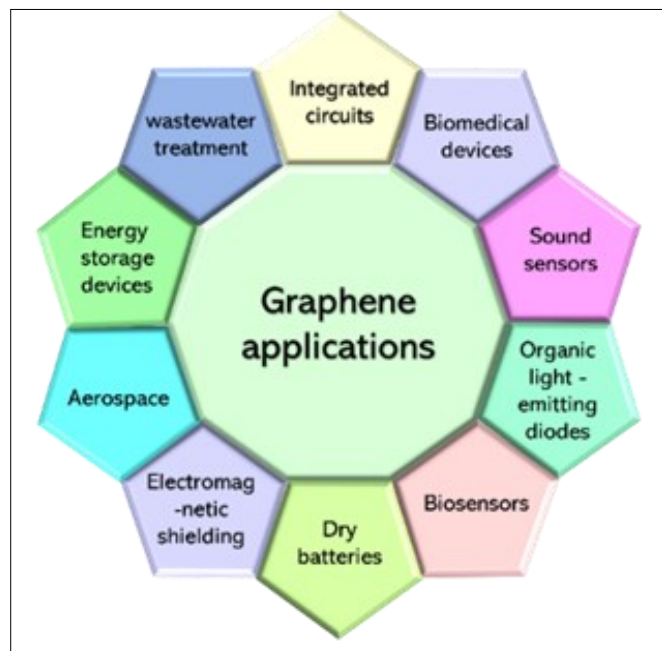


Fig. 2. Graphene applications in different sectors.

influence the stacking behaviour of graphene nanosheets. Structurally, it is a two-dimensional (2D) material with a planar architecture and an open surface. It may be considered analogous to a graphene nanoplatelet, enabling bidirectional adsorption (9). The chemical structure of the graphene is illustrated in Fig. 3.

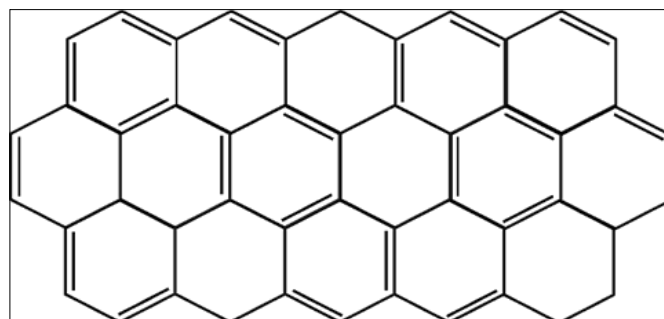


Fig. 3. Chemical structure of graphene.

Graphene based nanostructures

While numerous research endeavours are underway, the adsorption of environmental contaminants is exhibited by graphene, graphene oxide (GO), and their associated modified nanomaterials with exceptional performance (10).

Graphene oxide

Graphene oxide (GO) has a variety of functional groups that contain oxygen and a strong ability to remove ions and coloured chemicals (11). The GO is an atomic sheet of graphene with a hybrid structure made up of a blend of sp^2 and sp^3 hybridized carbon atoms with the thickness of one layer of graphene oxide nanosheet is ~ 0.8 to 1.0 nm (12, 13). It is an atomically thin layer 2D structure with many oxygenated functional groups on its basal planes and at its borders (14). The GO has high surface to weight ratio (theoretical value of $2620 \text{ m}^2 \text{ g}^{-1}$) and oxygenated groups including carboxyl ($-\text{COOH}$), epoxide ($-\text{COC}-$) and hydroxyl ($-\text{OH}$) groups (Fig. 4).

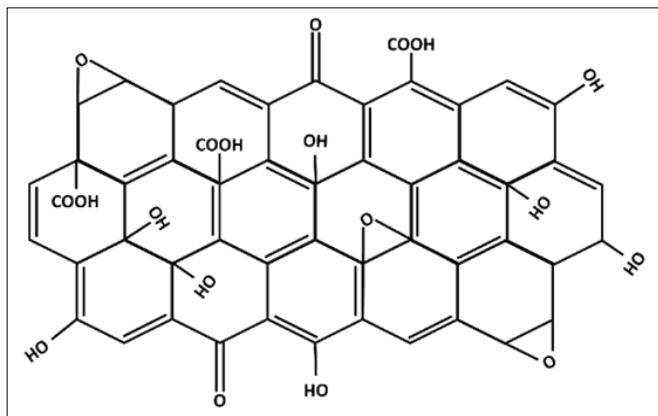


Fig. 4. Graphene oxide chemical structure contains hydroxyl, epoxy and carboxyl groups.

Small clusters of pure epoxy and -OH groups can be found within the carbon network (15). Additionally, graphene-like sections and gaps are frequently present in GO. These properties make GO hydrophilic, allowing it to be readily dissolved in water and other solvents (16). The GO was found to be composed of 2.588 % hydrogen and 47.665 % carbon with 50 % oxygen by weight, according to the element analysis results (17). Incorporation of nanoparticles such as ZIF-8, TiO_2 , BiOBr, $\text{g-C}_3\text{N}_4$ to GO membranes has strong hydrophilicity, permeability, anti-fouling and photocatalytic self-cleaning properties diverse applications efficient removal of dyes an organic pollutant in wastewater treatment (18, 20). The different methods used for the synthesis of GO is shown in Fig. 5.

Reduced graphene oxide (rGO)

Reduced graphene oxide (rGO), referred to as functionalised graphene, chemically modified graphene and chemically transformed graphene (Fig. 6) (19). The rGO developing into a very good graphene or GO compromise is also simple to synthesise appropriate amounts from inexpensive reduced GO utilising a range of (electro) chemical, microwave, photo-assisted and thermal reductions. Composite of TiO_2 , ZnO, SnO_2 and Ag nanoparticles exhibit enhances photocatalytic activity under visible light, leading to efficient degradation of dyes, pharmaceuticals and other organic contaminates (20, 21). The synergistics interaction of graphene's high electron mobility with the photocatalytic properties of metal oxides or noble metal minimize electron hole recombination, enhances light absorption and significantly improves degradation efficiency (22).

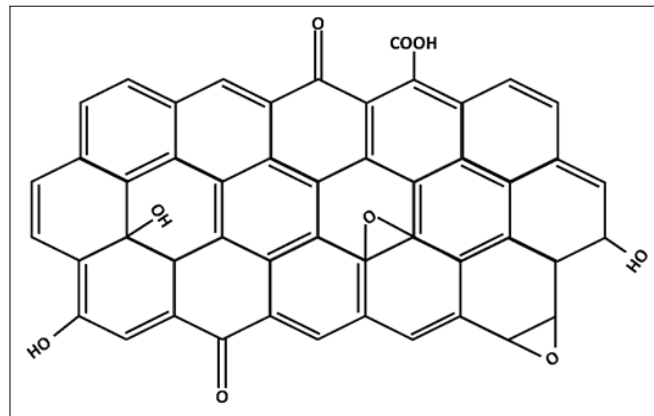


Fig. 6. Chemical structure of reduced graphene oxide.

Graphene nanoplatelets (GNPs)

Graphene nanoplatelets (GNPs), which offer potential for large-scale manufacturing at low cost, have emerged as an alternative. GNPs contain few graphite layers, ranging in thickness from 5 to 20 nm is shown in Fig. 7. This nanoflake powder can be produced without additional centrifugation steps through liquid-phase exfoliation. Ball milling, microwave radiation treatment of acid-intercalated graphite, shear exfoliation and contemporary wet-jet milling are other common processes (23, 24). The thickness, aspect ratio, lateral flake size and defect concentrations of the powders produced by these production processes vary greatly. The GNPs are architecturally in between graphene and graphite since they are made of one and a few layers of graphene combined with thicker graphite (25). Recent advances in chemical and mechanical exfoliation techniques enable high-yield, low-defect GNP production from graphite, facilitating their application in industrial wastewater treatment, particularly in corrosion resistance, sensing and advanced composite materials (26).

Graphene quantum d(GQDs)

Graphene quantum dots (GQDs) are zero-dimensional nanomaterials, typically less than 20 nm in diameter, that combine the unique properties of graphene with quantum confinement and edge effects. Application of GQDs, photo-induced electron transfer, tunable photoluminescence (PL), high solubility, minimal toxicity, and biocompatibility (27). Previously, carbon-based QDs had distinct advantages such as simple production, superior biocompatibility, nontoxicity, good metallic QD and physicochemical features like great stability and controllable fluorescence (28). Two-dimensional

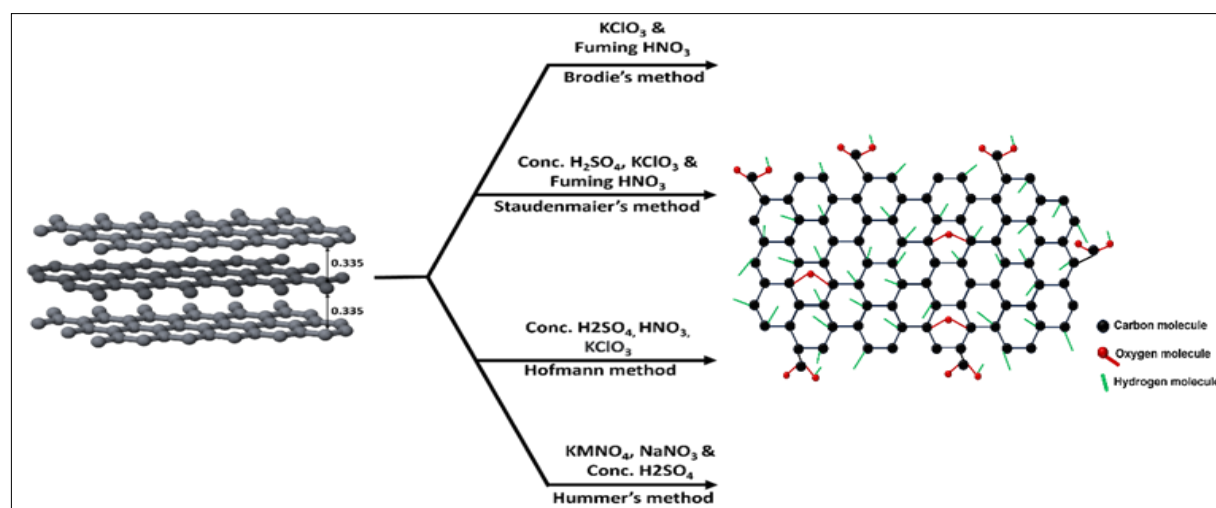


Fig. 5. Systematic diagram shows different methods for synthesis of graphene oxide.

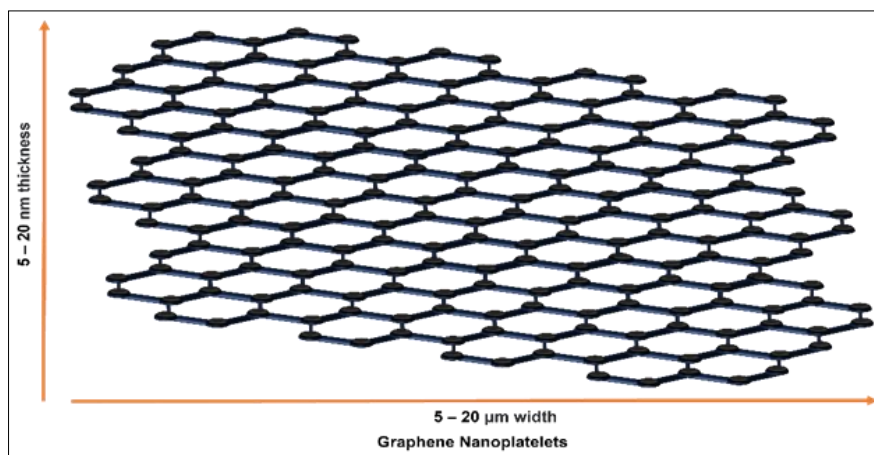


Fig. 7. Graphene nanoplatelets (GNPs).

graphene restricts electrical transport in all 3 spatial dimensions because of their graphene layer structure, graphene quantum dots (GQDs) exhibit superior chemical and physical properties over carbon nanodots. These properties include a larger length-to-diameter ratio, more surface area and stronger surface bonding with a double π - π network (29).

Graphene nanoribbons (GNRs)

Graphene nanoribbons (GNRs), which are less than 10 nm wide, have unique electrical and spin transport characteristics (30). The GNPs are in different forms based on carbon atom arrangement which are in zigzag and armchair patterns (Fig. 8a & b). But it's still practically difficult to produce high-quality nanoribbons in large quantities. To create graphene nanoribbons, lithographic, chemical and Sono-chemical techniques have been devised (31,32).

Graphene aerogels

Graphene aerogel (GA) or graphene foam has properties such as ultra-low density, high electrical conductivity, low thermal conductivity and super compressibility. The GA is especially attractive for practical applications since it may be attained in macroscopic monoliths with configurable hierarchical nanoarchitectures (33). GA can be produced by the sol-gel procedure, hydrothermal, chemical reduction and *in-situ* growth using chemical vapour deposition popularly known as chemical vapour deposition (CVD) (34-37).

Synthesis of graphene and its nano-derivatives

The synthesis of graphene and its above-stated nano-derivatives, such as graphene oxide, reduced graphene oxide, graphene nanoplatelets, graphene quantum dots, graphene nanoribbons and graphene aerogels, has gained significant attention due to their exceptional properties and wide application potential, with

synthesis methods advancing rapidly to improve scalability (Fig. 9). Top-down approaches such as mechanical exfoliation and liquid phase exfoliation are widely used to producing layer graphene with liquid phase methods often assisted by ultrasonication to enable scalable production (38). Oxidative exfoliation via the Hummers and modified Hummers methods convert graphite to GO, which can be reduced chemically, thermally or electrochemically into rGO, offering scalability and functional groups incorporation group incorporation to enhances dispersibility and reactivity (39).

Ultrasound assisted synthesis further accelerates exfoliation and functionalisation, producing thin layer GO or rGO with high yield and uniformity, while enabling homogeneous nanoparticle deposition a diverse graphene morphology such as foams and sponges (40). Bottom approaches such as CVD allow the growth of high quality, large area graphene films on metal substrates (Ni, Cu) and are extendable to the synthesis of GNRs and nanopores graphene through polymerization and cyclodehydrogenation of organic precursors, enabling control over width, edge structure and pore size (41). Green and biomass derived methods have recently emerged, enabling graphene synthesis from coal, biomass offering sustainable and cost-effective alternatives (42).

Functionalisation strategies, including covalent and non-covalent modifications such as esterification, imidation and surface decoration with metals or polymers, allow tailoring of GO and its derivative for specific application (43). Emulsion-based and *in situ* polymerization enhance graphene-polymer nanocomposite performance, while noble metal decoration broadens applications in catalysis and sensing. Industrially, oxidative exfoliation-reduction, liquid-phase exfoliation and CVD offer scalable synthesis, though challenges of environmental impact, aggregation and defect-free

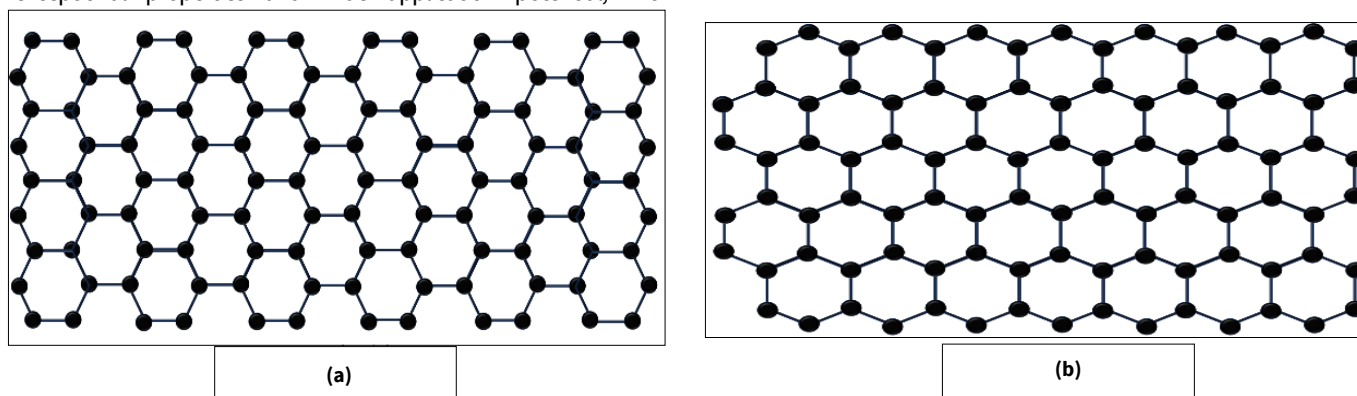


Fig. 8. Graphene nanoribbons. a: arm-chair; b: zig zag.

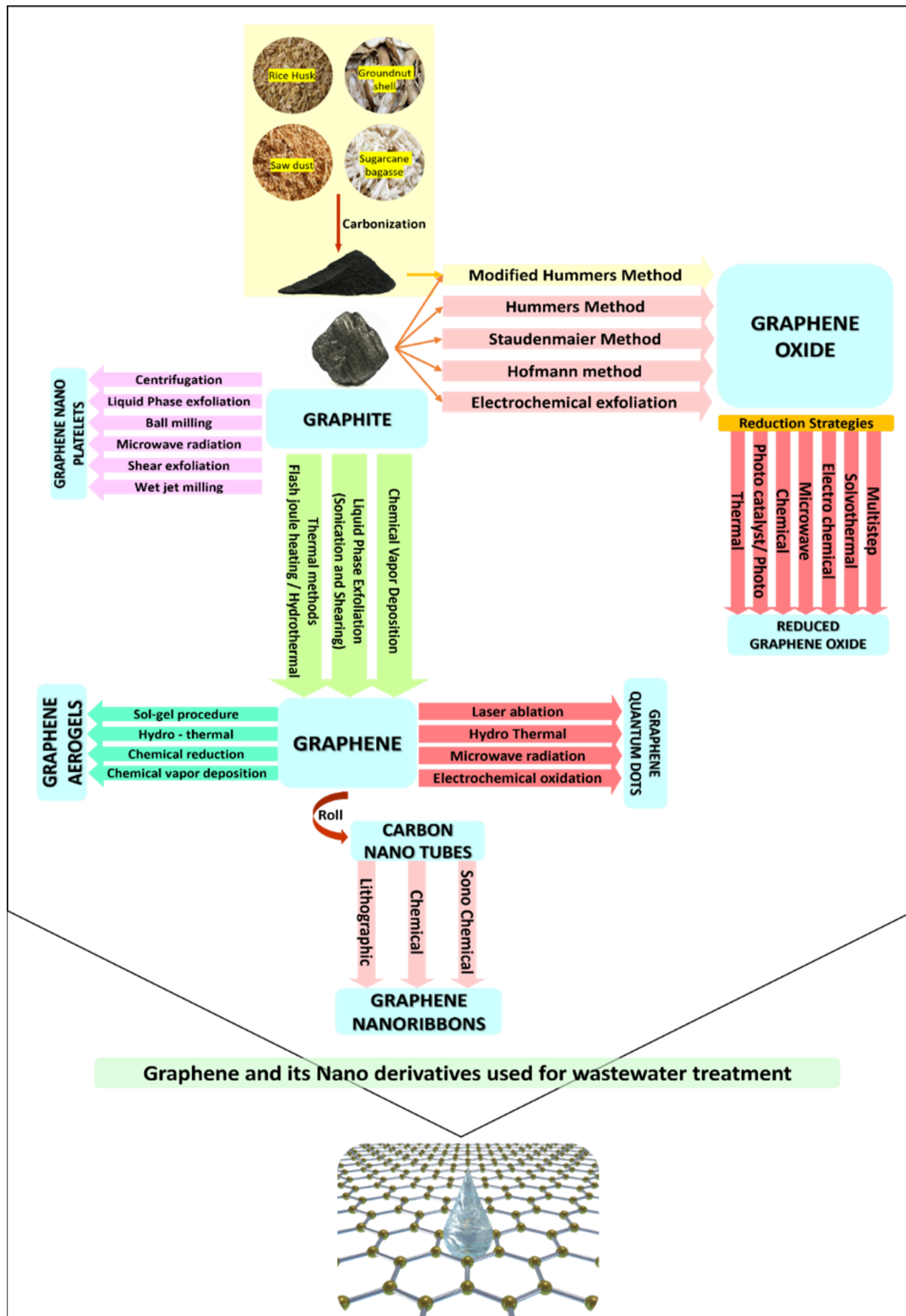


Fig. 9. Diagrammatic representation of various methods for the synthesis of graphene and its nano derivatives.

production persist (44). Overall, graphene synthesis advances toward sustainable, scalable and application-specific strategies.

Biodegradable graphene and different graphene nano composites used in wastewater treatment

Biodegradable carbon nanoparticles provide great potential for transforming wastewater treatment procedures and providing an efficient, long-term and low-cost solution to environmental issues. These nanoparticles have unique characteristics such as large surface area, non-toxicity, adjustable porosity and are incredibly effective at adsorbing and catalysing the degradation of a wide range of pollutants, from organic contaminants to heavy metals, in the context of wastewater treatment (45, 46). The ability of naphthalene-degrading bacteria to oxidise and degrade the graphitic materials including GO, graphite and reduced GO by the process of oxidation (47). The interactions between lignin peroxidase (LiP), a ligninolytic enzyme secreted from the white rot fungus (*Phanerochaete chrysosporium*) and oxidised and reduced graphene oxide nanoribbons (GONRs and rGONRs) (48). The composite was highly crystalline, thermally stable and exhibited excellent surface activity, making it a sustainable alternative to conventional filters.

Alginate-coated iron oxide GO composites, an eco-friendly material, exhibit > 90 % removal efficiency for dyes such as methylene blue and malachite green through adsorption and photocatalytic degradation, while maintaining strong reusability over multiple cycles (49). Xanthan biopolymer/GO-TiO₂ 3D composites exhibit high lead adsorption capacity (199 mg g⁻¹) and excellent recyclability, retaining over 84 % efficiency after five cycles (50).

Graphene-based composites have demonstrated remarkable potential for pollutant removal due to their high adsorption capacity, photocatalytic activity and recyclability. GO-cellulose nanocrystal membranes can eliminate up to 99 % of heavy metals and dyes with high sustainability, while alginate-coated Fe₃O₄ at GO achieves more than 90 % dye removal and maintains over 85 % efficiency after 6 regeneration cycles (49, 51). Bio-graphene-magnetite composites exhibit high adsorption capacity (>180 mg g⁻¹) for dyes and Cr⁶⁺, with the added advantage of magnetic recovery (52). Similarly, xanthan/GO-TiO₂ 3D composites remove lead ions up to 199 mg g⁻¹ with >84 % efficiency after five cycles, ensuring long-term usability (53). Advanced hybrids such as TiO₂/GO, ZnFe₂O₄/graphene and LDH/rGO achieve nearly 100 % removal of dyes, antibiotics and other organic contaminants, with biodegradability and regeneration capacity depending on the composite matrix (54).

A comprehensive summary of recent studies that explore the application of graphene nano-derivatives in wastewater treatment are given in Table 1. The table highlights the various modifications of graphene materials, method of synthesis, the pollutants targeted and the mechanism of adsorption. This synthesis offers valuable insights into the versatility of graphene-based adsorbents in addressing a wide range of contaminants, illustrating both their effectiveness and potential challenges. By consolidating these findings, the table serves as a reference point for future research, helping to identify trends and gaps in the field.

Future developments and applications

The application of nanotechnology in the treatment of water and wastewater is becoming inevitable due to its constant evolution. The present water purification technologies are thought to be adequately replaced by 2D nanoporous materials due to their outstanding mechanical strength, high surface areas, and atomically

thin architectures. The production of materials based on nanoporous graphene opened up a world of possibilities for water purification technology. Inorganic nano-adsorbent nanoparticles are utilised to remove pollutants from wastewater because of their wonderful, anticipated qualities that rely on their structure (73). To utilise nanomaterial as a sorbent, some traits such as nontoxic, high sorption capacity and indefinitely reusable by removing toxins from the surface of the nanomaterial are crucial.

Functionalization methods might modify graphene's surface characteristics to maximise how well it interacts with contaminants. The creation of graphene-based nanocomposites may improve adsorption even more and make it possible to remove several pollutants at once (74, 75). Furthermore, scalable fabrication technology developments may make it easier to produce graphene membranes on a wide scale, increasing their suitability for real-world uses in wastewater treatment facilities. The breakdown of organic contaminants in wastewater is mostly dependent on advanced oxidation processes (AOPs), where graphene-based nanomaterials have exceptional catalytic activity (76). Electrodes based on graphene have demonstrated potential in electrochemical methods for treating wastewater, such as capacitive deionization, electrooxidation and electrocoagulation. Tailored nanocomposite materials such as GQDs, GONRs and GA are used for wastewater treatment.

Conclusion

Graphene and graphene-based nanomaterials such as GO, rGO, GQD, GONR and GA provide flexible ways to tackle the growing problems associated with water pollution and bring a paradigm change in the field of wastewater treatment. A wide range of applications, from adsorption and membrane filtration to catalysis and sensing, are made possible by their remarkable features, which contain mechanical strength, large surface area and catalytic activity. The full potential of graphene-based nanomaterials in wastewater treatment is not identified and this needs novel investigation in nanocomposite designs, synthesis methods and integrating them into useful treatment systems. Multidisciplinary and cross-sector collaboration is essential to convert laboratory discoveries into globally applicable, sustainable solutions for resource recovery and water purification. By making use of graphene's extraordinary qualities, we can clear the path for a more sustainable and clean future for water resources.

Acknowledgements

The authors express their gratitude to Tamil Nadu Agricultural University for providing the necessary resources and research facilities for this study.

Authors' contributions

PPC, MTGR and RK¹ conceptualised the study, designed the methodology and supervised the research. MTGR, PPC performed the literature review and data analysis. SP, KG contributed to the manuscript writing and figure preparation. RK¹, KR, RK², NRM and RSS conducted the technical validation and reviewed the final draft. MTGR, AJJ, KG assisted in manuscript editing and formatting. All authors read and approved the final manuscript. [RK¹ stands for Raja Kalimuthu and RK² stands for Raghunath Kaliaperumal].

Table 1. Graphene nano derivatives used in wastewater treatment

S. no.	Adsorbent	Target contaminant	Method of synthesis	Adsorption Capacity (milligram per gram - mg g ⁻¹ / percentage - %)	Mechanism of adsorption	References
Graphene oxide						
1	Granular activated carbon (GAC) - graphene oxide (GO)	Chloroquine and dipyrone	Modified Hummers method	37.65 mg g ⁻¹ and 62.43 mg g ⁻¹ for chloroquine and dipyrone adsorption respectively	π-π interactions	(55)
2	Persulfate activated magnetic Fe ₃ O ₄ /graphene oxide nanocomposite	Malathion	Modified Hummers method	96.5%	π-π interactions	(56)
3	Graphene oxide and agar biopolymer (agar-GO)	Nile Blue, Methylene Blue, Malachite Green, Basic Fuschin	Modified Hummers method	226.46 mg·g ⁻¹ (Nile Blue), 79.51 mg·g ⁻¹ (Methylene Blue), 58.25 mg·g ⁻¹ (Malachite Green), 38.11 mg·g ⁻¹ (Basic Fuschin)	π-π interactions and electrostatic interactions	(57)
4	Graphene - CdO and graphene oxide - CdO nanocomposites	Cr (VI)	Electrochemical exfoliation method	699 mg g ⁻¹	Electrostatic interactions	(58)
Reduced graphene oxide						
5	Reduced graphene oxide decorated with zinc ferrite nanoparticles	Pb (II) and MB dye	Doping of ZnFe ₂ O ₄ into rGO	Pb (II) and MB dye of 89.8 and 119.0 mg g ⁻¹ , respectively	π-π interactions	(59)
6	Reduced graphene oxide nanosheets coated with ZnS and ZnS-Ag	Malachite green (MG) and Ethyl violet (EV)	rGO-ZnS doped by Ag	25.72 mg g ⁻¹ for MG and 18.34 mg g ⁻¹ for EV	Electrostatic interactions	(60)
7	Montmorillonite - reduced graphene oxide - PEG polymer	Enrofloxacin	Thermal reduction process	310.6 mg g ⁻¹	π-π interactions	(61)
8	Reduced graphene oxide	Phenols	Sonication method	994 mg g ⁻¹	π-π interactions	(62)
Graphene nanoplatelets						
9	Graphene Nanoplatelets	As (III)	<i>In-situ</i> methodology by using ferric chloride	~99% at 300 K temperature	Ion- dipole interaction	(63)
10	Graphene nanoplatelets/ Polythiophene nanocomposites	Bromo phenol blue (BPP) and Pathogen control	<i>In-situ</i> Oxidative chemical polymerization technique	94.1%	Electrostatic attraction	(64)
11	SA-GNP (sodium alginate-graphene nanoplatelet composite) beads	Methylene blue (MB)	Gelatinization process	290.36 mg g ⁻¹	Chemisorption	(65)
Graphene quantum dots						
12	Fe ₂ O ₃ -GQD/NF-TiO ₂	Cr (VI)	Hydrothermal method	935 mg g ⁻¹	π-π interactions	(66)
13	GQDs-TiO ₂ /AC	Methylene blue (MB)	Microwave assisted preparation	313 mg g ⁻¹	π-π interactions	(67)
Graphene oxide nanoribbons						
14	Aerogels made of sodium alginate, graphene oxide, and graphene oxide nanoribbons (GO/GONRs/SA)	U (VI)	Modified Hummer's process followed by freeze-drying	929.16 mg g ⁻¹	Electrostatic and π-π interactions	(68)
15	Graphene oxide nanoribbons (GONRs)	As (III) and Hg (II)	Oxidative unzipping of multi walled CNTs to GONRs	As(V) - 155.61 mg g ⁻¹ Hg(II) - 33.02 mg g ⁻¹	π-π interactions	(69)
Graphene and graphene oxide aerogels						
16	Nanocellulose and graphene oxide aerogels (CNF/GO)	Methyl blue (MB)	-	99% in 20 minutes	π-π interactions	(70)
17	Lysine and EDA double cross-linked graphene aerogel (LEGA)	Methylene blue (MB)	Ultrasonication method	332.23 mg g ⁻¹	π-π interactions	(71)
18	N-doped graphene oxide aerogel	Phenols	-	108.19 mg g ⁻¹	π-π interactions	(72)

Compliance with ethical standards

Conflict of interest: Authors do not have any conflict of interests to declare.

Ethical issues: None

References

- Li Y, Kontos GA, Cabrera DV, Avila NM, Parkinson TW, Viswanathan MB, et al. Design of a high-rate wastewater treatment process for energy and water recovery at biorefineries. *ACS Sustain Chem Eng*. 2023;11:3861-72. <https://doi.org/10.1021/acssuschemeng.2c07139>
- Thakur A, Kumar A, Singh A. Adsorptive removal of heavy metals, dyes and pharmaceuticals: Carbon-based nanomaterials in focus. *Carbon*. 2024;217:118621. <https://doi.org/10.1016/j.carbon.2023.118621>
- Khalid ZB, Nasrullah M, Nayeem A, Wahid ZA, Singh L, Krishnan S. Application of 2D graphene-based nanomaterials for pollutant removal from advanced water and wastewater treatment processes. In: *Adapting 2D nanomaterials for advanced applications*. Washington (DC): ACS Publications; 2020. p. 191-217. <https://doi.org/10.1021/bk-2020-1353.ch009>
- Wang Z, Shen D, Wu C, Gu S. State-of-the-art on the production and application of carbon nanomaterials from biomass. *Green Chem*. 2018;20:5031-57. <https://doi.org/10.1039/C8GC01748D>
- Penev ES, Artyukhov VI, Ding F, Yakobson BI. Unfolding the fullerene: Nanotubes, graphene and poly-elemental varieties by simulations. *Adv Mater*. 2012;24:4956-76. <https://doi.org/10.1002/adma.201202322>
- Geim AK. Graphene: Status and prospects. *Science*. 2009;324:1530-34. <https://doi.org/10.1126/science.1158877>
- Stankovich S, Dikin DA, Piner RD, Kohlhaas KA, Kleinhammes A, Jia Y, et al. Synthesis of graphene-based nanosheets via chemical reduction

- of exfoliated graphite oxide. *Carbon*. 2007;45:1558-65. <https://doi.org/10.1016/j.carbon.2007.02.034>
8. Han TH, Kim H, Kwon SJ, Lee TW. Graphene-based flexible electronic devices. *Mater Sci Eng R Rep*. 2017;118:1-43. <https://doi.org/10.1016/j.mser.2017.05.001>
 9. Jayakaran P, Nirmala GS, Govindarajan L. Qualitative and quantitative analysis of graphene-based adsorbents in wastewater treatment. *Int J Chem Eng*. 2019;2019:1-17. <https://doi.org/10.1155/2019/9872502>
 10. Yin Z, Cui C, Chen H, Duoni, Yu X, Qian W. The application of carbon nanotube/graphene-based nanomaterials in wastewater treatment. *Small*. 2020;16:1902301. <https://doi.org/10.1002/smll.201902301>
 11. Bello MM, Raman AAA. Adsorption and oxidation techniques to remove organic pollutants from water. In: *Green adsorbents for pollutant removal: Fundamentals and design*. Cham: Springer; 2018. p. 249-300. https://doi.org/10.1007/978-3-319-92111-2_8
 12. Srivastava SK, Pionteck J. Recent advances in preparation, structure, properties and applications of graphite oxide. *J Nanosci Nanotechnol*. 2015;15:1984-2000. <https://doi.org/10.1166/jnn.2015.10047>
 13. Ren X, Li J, Tan X, Wang X. Comparative study of graphene oxide, activated carbon and carbon nanotubes as adsorbents for copper decontamination. *Dalton Trans*. 2013;42:5266-74. <https://doi.org/10.1039/c3dt32969k>
 14. Schafhaeuti C. On the combinations of carbon with silicon and iron, and other metals, forming the different species of cast iron, steel, and malleable iron. *Lond Edinb Dubl Philos Mag J Sci*. 1840;16:570-90. <https://doi.org/10.1080/1478644008650094>
 15. Brodie BC. On the atomic weight of graphite. *Philos Trans R Soc Lond*. 1859;149:249-59. <https://doi.org/10.1098/rstl.1859.0013>
 16. Zhang J, Lu X, Shi C, Yan B, Gong L, Chen J, et al. Unraveling the molecular interaction mechanism between graphene oxide and aromatic organic compounds with implications on wastewater treatment. *Chem Eng J*. 2019;358:842-49. <https://doi.org/10.1016/j.cej.2018.10.064>
 17. Mahmoodi NM, Maroofi SM, Mazarji M, Nabi-Bidhendi G. Preparation of modified reduced graphene oxide nanosheet with cationic surfactant and its dye adsorption ability from colored wastewater. *J Surfactants Deterg*. 2017;20:1085-93. <https://doi.org/10.1007/s11743-017-1985-1>
 18. Long Q, Chen L, Zong Y, Wan X, Liu F, Luo H, et al. Photocatalytically self-cleaning graphene oxide nanofiltration membranes reinforced with bismuth oxybromide for high-performance water purification. *J Colloid Interface Sci*. 2024;675:958-69. <https://doi.org/10.1016/j.jcis.2024.07.027>
 19. Kyzas GZ, Deliyanni EA, Matis KA. Graphene oxide and its application as an adsorbent for wastewater treatment. *J Chem Technol Biotechnol*. 2014;89:196-205. <https://doi.org/10.1002/jctb.4220>
 20. Mousa SA, Abdallah H, Ibrahim SS, Khairy SA. Enhanced photocatalytic properties of graphene oxide/polyvinylchloride membranes by incorporation with green prepared SnO₂ and TiO₂ nanocomposite for water treatment. *Appl Phys A*. 2023;129:831. <https://doi.org/10.1007/s00339-023-07117-8>
 21. Yu Z, Zeng H, Min X, Zhu X. High-performance composite photocatalytic membrane based on titanium dioxide nanowire/graphene oxide for water treatment. *J Appl Polym Sci*. 2020;137:48488. <https://doi.org/10.1002/app.48488>
 22. Pei S, Cheng HM. The reduction of graphene oxide. *Carbon*. 2012;50:3210-28. <https://doi.org/10.1016/j.carbon.2011.11.010>
 23. An SJ, Zhu Y, Lee SH, Stoller MD, Emilsson T, Park S, et al. Thin film fabrication and simultaneous anodic reduction of deposited graphene oxide platelets by electrophoretic deposition. *J Phys Chem Lett*. 2010;1:1259-63. <https://doi.org/10.1021/jz100080c>
 24. Zhu Y, Murali S, Stoller MD, Velamakanni A, Piner RD, Ruoff RS. Microwave assisted exfoliation and reduction of graphite oxide for ultracapacitors. *Carbon*. 2010;48:2118-22. <https://doi.org/10.1016/j.carbon.2010.02.001>
 25. Im MJ, Kim JI, Hyeong SK, Moon BJ, Bae S. From pristine to heteroatom-doped graphene quantum dots: An essential review and prospects for future research. *Small*. 2023;19:2304497. <https://doi.org/10.1002/smll.202304497>
 26. Ghadami Karder H, Pircheraghi G. Polyethylene/graphene-nanoplatelets nanocomposites with improved thermochemical stability: The role of surface polarity of graphene-nanoplatelets. *Polym Compos*. 2024;45:16262-79. <https://doi.org/10.1002/pc.28904>
 27. Wu ZS, Ren W, Gao L, Liu B, Jiang C, Cheng HM. Synthesis of high-quality graphene with a pre-determined number of layers. *Carbon*. 2009;47:493-99. <https://doi.org/10.1016/j.carbon.2008.10.031>
 28. Young RJ, Kinloch IA, Gong L, Novoselov KS. The mechanics of graphene nanocomposites: A review. *Compos Sci Technol*. 2012;72:1459-76. <https://doi.org/10.1016/j.compscitech.2012.05.005>
 29. Del Rio Castillo AE, Pellegrini V, Ansaldo A, Ricciardella F, Sun H, Marasco L, et al. High-yield production of 2D crystals by wet-jet milling. *Mater Horiz*. 2018;5:890-904. <https://doi.org/10.1039/C8MH00487K>
 30. Wick P, Louw-Gaume AE, Kucki M, Krug HF, Kostarelos K, Fadeel B, et al. Classification framework for graphene-based materials. *Angew Chem Int Ed Engl*. 2014;53:7714-18. <https://doi.org/10.1002/anie.201403335>
 31. Zhou S, Xu H, Gan W, Yuan Q. Graphene quantum dots: Recent progress in preparation and fluorescence sensing applications. *RSC Adv*. 2016;6:110775-88. <https://doi.org/10.1039/C6RA24349E>
 32. Guan X, Li Z, Geng X, Lei Z, Karakoti A, Wu T, et al. Emerging trends of carbon-based quantum dots: Nanoarchitectonics and applications. *Small*. 2023;19:2207181. <https://doi.org/10.1002/smll.202207181>
 33. Dorončić S, Jovanović S, Bonasera A. Shedding light on graphene quantum dots: Key synthetic strategies, characterization tools, and cutting-edge applications. *Materials*. 2021;14:6153. <https://doi.org/10.3390/ma14206153>
 34. Jeong S, Pinals RL, Dharmadhikari B, Song H, Kalluri A, Debnath D, et al. Graphene quantum dot oxidation governs noncovalent biopolymer adsorption. *Sci Rep*. 2020;10:7074. <https://doi.org/10.1038/s41598-020-63769-z>
 35. Karkooti A, Yazdi AZ, Chen P, McGregor M, Nazemifard N, Sadrzadeh M. Development of advanced nanocomposite membranes using graphene nanoribbons and nanosheets for water treatment. *J Membr Sci*. 2018;560:97-107. <https://doi.org/10.1016/j.memsci.2018.04.034>
 36. Chen Z, Lin YM, Rooks MJ, Avouris P. Graphene nano-ribbon electronics. *Physica E*. 2007;40:228-32. <https://doi.org/10.1016/j.physe.2007.06.020>
 37. Campos LC, Manfrinato VR, Sanchez-Yamagishi JD, Kong J, Jarillo-Herrero P. Anisotropic etching and nanoribbon formation in single-layer graphene. *Nano Lett*. 2009;9:2600-04. <https://doi.org/10.1021/nl900811r>
 38. Ashok Kumar SS, Bashir S, Ramesh K, Ramesh S. A review on graphene and its derivatives as the forerunner of the two-dimensional material family for the future. *J Mater Sci*. 2022;57:12236-78. <https://doi.org/10.1007/s10853-022-07346-x>
 39. Singh S, Hasan MR, Sharma P, Narang J. Graphene nanomaterials: The wondering material from synthesis to applications. *Sensors (Basel)*. 2022;3:100190. <https://doi.org/10.1016/j.sintl.2022.100190>
 40. Muthoosamy K, Manickam S. State of the art and recent advances in the ultrasound-assisted synthesis, exfoliation and functionalization of graphene derivatives. *Ultrason Sonochem*. 2017;39:478-93. <https://doi.org/10.1016/j.ultsonch.2017.05.019>
 41. Qin T, Wang T, Zhu J. Recent progress in on-surface synthesis of nanoporous graphene materials. *Commun Chem*. 2024;7:154. <https://doi.org/10.1038/s42004-024-01222-2>
 42. Tamuly J, Bhattacharjya D, Saikia BK. Graphene/graphene derivatives from coal, biomass, and wastes: Synthesis, energy applications, and perspectives. *Energy Fuels*. 2022;36:12847-74. <https://doi.org/10.1021/acs.energyfuels.2c00976>
 43. Gulia S, Moniruzzaman M, Panda A. Approaches in graphene-based nanocomposites: Synthesis, modification, and multifaceted applications. *FlatChem*. 2024;48:100761. <https://doi.org/10.1016/j.flatc.2024.100761>
 44. Backes C, Abdelkader AM, Alonso C, Andrieux-Ledier A, Arenal R, Azpeitia J, et al. Production and processing of graphene and related materials. *2D Mater*. 2020;7:022001. <https://doi.org/10.1088/2053-1583/ab1e0a>

45. Li D, Müller MB, Gilje S, Kaner RB, Wallace GG. Processable aqueous dispersions of graphene nanosheets. *Nat Nanotechnol.* 2008;3:101-05. <https://doi.org/10.1038/nnano.2007.451>
46. Prabu PC. Impact of heavy metal contamination of Akaki River of Ethiopia on soil and metal toxicity on cultivated vegetable crops. *Electron J Environ Agric Food Chem.* 2009;8:18-27.
47. Liu L, Zhu C, Fan M, Chen C, Huang Y, Hao Q, et al. Oxidation and degradation of graphitic materials by naphthalene-degrading bacteria. *Nanoscale.* 2015;7:13619-28. <https://doi.org/10.1039/C5NR02502H>
48. Xu Z, Zhang Y, Li P, Gao C. Strong, conductive, lightweight, neat graphene aerogel fibers with aligned pores. *ACS Nano.* 2012;6:7103-13. <https://doi.org/10.1021/nn3021772>
49. Mubarak MF, Adaileh A, Ahmed IA, Ali MA, Hemdan M. Eco-friendly alginate-coated nano iron oxide-graphene oxide nanocomposite for high-performance adsorption and photocatalytic detoxification of harmful dyes in wastewater treatment. *J Cluster Sci.* 2025;36:27. <https://doi.org/10.1007/s10876-024-02740-8>
50. Hu H, Wen W, Ou JZ. Construction of adsorbents with graphene and its derivatives for wastewater treatment: A review. *Environ Sci Nano.* 2022;9:3226-76. <https://doi.org/10.1039/D2EN00248E>
51. Sheikh MS, Rahman MM, Rahman MS, Yildirim K, Maniruzzaman M. Fabrication of nano composite membrane filter from graphene oxide and banana rachis cellulose nano crystal for industrial effluent treatment. *J Ind Eng Chem.* 2023;128:196-208. <https://doi.org/10.1016/j.jiec.2023.07.048>
52. Ramalingam B, Parandhaman T, Choudhary P, Das SK. Biomaterial functionalized graphene-magnetite nanocomposite: A novel approach for simultaneous removal of anionic dyes and heavy-metal ions. *ACS Sustain Chem Eng.* 2018;6:6328-41. <https://doi.org/10.1021/acssuschemeng.8b00139>
53. Lim MB, Hu M, Manandhar S, Sakshaug A, Strong A, Riley L, et al. Ultrafast sol-gel synthesis of graphene aerogel materials. *Carbon.* 2015;95:616-24. <https://doi.org/10.1016/j.carbon.2015.08.037>
54. Shan H, Xiong D, Li X, Sun Y, Yan B, Li D, et al. Tailored lithium storage performance of graphene aerogel anodes with controlled surface defects for lithium-ion batteries. *Appl Surf Sci.* 2016;364:651-59. <https://doi.org/10.1016/j.apsusc.2015.12.143>
55. Chen M, Li Z, Li J, Li J, Li Q, Zhang L. The extraction of uranium using graphene aerogel loading organic solution. *Talanta.* 2017;166:284-91. <https://doi.org/10.1016/j.talanta.2017.01.070>
56. Shi L, Chen K, Du R, Bachmatiuk A, Rummeli MH, Xie K, et al. Scalable seashell-based chemical vapor deposition growth of three-dimensional graphene foams for oil-water separation. *J Am Chem Soc.* 2016;138:6360-63. <https://doi.org/10.1021/jacs.6b02262>
57. Januário EFD, Fachina YJ, Wernke G, Demiti GMM, Beltran LB, Bergamasco R, et al. Application of activated carbon functionalized with graphene oxide for efficient removal of COVID-19 treatment-related pharmaceuticals from water. *Chemosphere.* 2022;289:133213. <https://doi.org/10.1016/j.chemosphere.2021.133213>
58. Dolatabadi M, Świergosz T, Wang C, Ahmadzadeh S. Accelerated degradation of groundwater-containing malathion using persulfate activated magnetic Fe₃O₄/graphene oxide nanocomposite for advanced water treatment. *Arab J Chem.* 2023;16:104424. <https://doi.org/10.1016/j.arabjc.2022.104424>
59. De Araujo CMB, Ghislandi MG, Rios AG, da Costa GRB, do Nascimento BF, Ferreira AFP, et al. Wastewater treatment using recyclable agar-graphene oxide biocomposite hydrogel in batch and fixed-bed adsorption column: Bench experiments and modeling for the selective removal of organics. *Colloids Surf A Physicochem Eng Asp.* 2022;639:128357. <https://doi.org/10.1016/j.colsurfa.2022.128357>
60. Bukhari A, Ijaz I, Zain H, Mehmood U, Iqbal MM, Gilani E, et al. Introduction of CdO nanoparticles into graphene and graphene oxide nanosheets for increasing adsorption capacity of Cr from wastewater collected from petroleum refinery. *Arab J Chem.* 2023;16:104445. <https://doi.org/10.1016/j.arabjc.2022.104445>
61. Ahmed MA, Ahmed MA, Mohamed AA. Facile adsorptive removal of dyes and heavy metals from wastewaters using magnetic nanocomposite of zinc ferrite@reduced graphene oxide. *Inorg Chem Commun.* 2022;144:109912. <https://doi.org/10.1016/j.inoche.2022.109912>
62. Naeem H, Tofil HM, Soliman M, Hai A, Zaidi SHH, Kizilbash N, et al. Reduced graphene oxide-zinc sulfide nanocomposite decorated with silver nanoparticles for wastewater treatment by adsorption, photocatalysis and antimicrobial action. *Molecules.* 2023;28:926. <https://doi.org/10.3390/molecules28030926>
63. Akpotu SO, Diagboya PN, Lawal IA, Sanni SO, Pholosi A, Peleyeju MG, et al. Designer composite of montmorillonite-reduced graphene oxide-PEG polymer for water treatment: Enrofloxacin sequestration and cost analysis. *Chem Eng J.* 2023;453:139771. <https://doi.org/10.1016/j.cej.2022.139771>
64. Haydari I, Aziz K, Kaya S, Daştan T, Ouazzani N, Mandi L, et al. Green synthesis of reduced graphene oxide and their use on column adsorption of phenol from olive mill wastewater. *Process Saf Environ Prot.* 2023;170:1079-91. <https://doi.org/10.1016/j.psep.2022.12.086>
65. Nandi D, Ghosh SK, Ghosh A, Siengchin S, Roy A, Gupta K, et al. Arsenic removal from water by graphene nanoplatelets prepared from nail waste: A physicochemical study of adsorption based on process optimization, kinetics, isotherm and thermodynamics. *Environ Nanotechnol Monit Manag.* 2021;16:100564. <https://doi.org/10.1016/j.enmm.2021.100564>
66. Noreen H, Iqbal J, Hassan W, Rahman G, Yaseen M, Rahman AU. Synthesis of graphene nanoplatelets/polythiophene nanocomposites with enhanced photocatalytic degradation of bromophenol blue and antibacterial properties. *Mater Res Bull.* 2021;142:111435. <https://doi.org/10.1016/j.materresbull.2021.111435>
67. Civan Çavuşoğlu F. Synthesis of graphene nanoplatelet-alginate composite beads and removal of methylene blue from aqueous solutions. *J Turk Chem Soc A.* 2023;10:287-302. <https://doi.org/10.18596/jotcsa.1196282>
68. Nagaraj A, Munusamy MA, Al-Arfaj AA, Rajan M. Functional ionic liquid-capped graphene quantum dots for chromium removal from chromium contaminated water. *J Chem Eng Data.* 2019;64:651-67. <https://doi.org/10.1021/acs.jced.8b00887>
69. Tohamy HAS, Fathy NA, El-Sakhawy M, Kamel S. Boosting the adsorption capacity and photocatalytic activity of activated carbon by graphene quantum dots and titanium dioxide. *Diam Relat Mater.* 2023;132:109640. <https://doi.org/10.1016/j.diamond.2022.109640>
70. Jabbar AA, Hussain DH, Latif KH, Albukhaty S, Jasim AK, Sulaiman GM, et al. Extremely efficient aerogels of graphene oxide/graphene oxide nanoribbons/sodium alginate for uranium removal from wastewater solution. *Sci Rep.* 2024;14:1285. <https://doi.org/10.1038/s41598-024-52043-1>
71. Dimiev AM, Khannanov A, Vakhitov I, Kiamov A, Shukhina K, Tour JM. Revisiting the mechanism of oxidative unzipping of multiwall carbon nanotubes to graphene nanoribbons. *ACS Nano.* 2018;12:3985-93. <https://doi.org/10.1021/acsnano.8b01617>
72. Sadeghi MH, Tofighy MA, Mohammadi T. One-dimensional graphene for efficient aqueous heavy metal adsorption. *Chemosphere.* 2020;253:126647. <https://doi.org/10.1016/j.chemosphere.2020.126647>
73. Nguyen VT, Ha LQ, Nguyen TDL, Ly PH, Nguyen DM, Hoang D. Nanocellulose and graphene oxide aerogels for adsorption and removal of methylene blue from an aqueous environment. *ACS Omega.* 2022;7:1003-13. <https://doi.org/10.1021/acsomega.1c05586>
74. Jiang L, Wen Y, Zhu Z, Liu X, Shao W. A double cross-linked strategy to construct graphene aerogels with highly efficient methylene blue adsorption performance. *Chemosphere.* 2021;265:129169. <https://doi.org/10.1016/j.chemosphere.2020.129169>
75. Cui Y, Kang W, Hu J. Effectiveness and mechanisms of the adsorption of phenol from wastewater onto N-doped graphene oxide aerogel. *J Water Process Eng.* 2023;53:103665. <https://doi.org/10.1016/j.jwpe.2023.103665>
76. Subramanian P, Pandian K, Pakkiam S, Dhanuskodi KV, Annamalai S, Padanillay Chidambaram P, et al. Biochar for heavy metal cleanup in soil and water: A review. *Biomass Convers Biorefin.* 2024;1:21. <https://doi.org/10.1007/s13399-024-05989-1>

Additional information

Peer review: Publisher thanks Sectional Editor and the other anonymous reviewers for their contribution to the peer review of this work.

Reprints & permissions information is available at https://horizonpublishing.com/journals/index.php/PST/open_access_policy

Publisher's Note: Horizon e-Publishing Group remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Indexing: Plant Science Today, published by Horizon e-Publishing Group, is covered by Scopus, Web of Science, BIOSIS Previews, Clarivate Analytics, NAAS, UGC Care, etc

See https://horizonpublishing.com/journals/index.php/PST/indexing_abstracting

Copyright: © The Author(s). This is an open-access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution and reproduction in any medium, provided the original author and source are credited (<https://creativecommons.org/licenses/by/4.0/>)

Publisher information: Plant Science Today is published by HORIZON e-Publishing Group with support from Empirion Publishers Private Limited, Thiruvananthapuram, India.